7.3.2 Gauss-Seidel iteration

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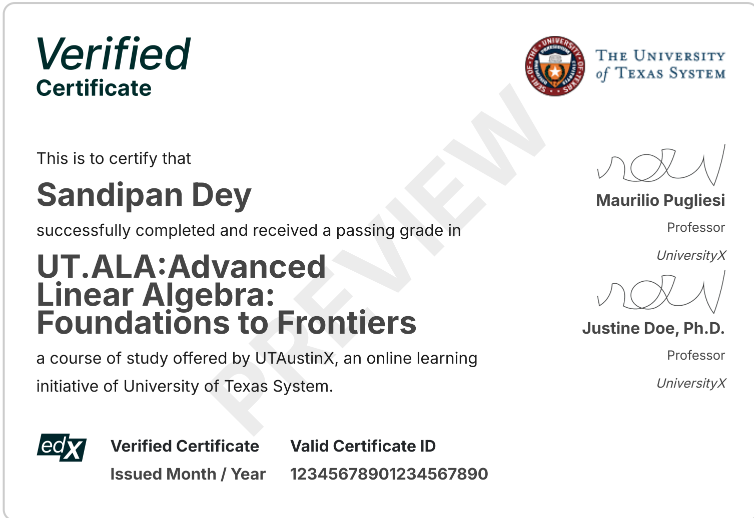

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Now what does that mean?

What that means is that our splitting here

now becomes ... where M is the diagonal (plus the strictly lowered, sorry) minus the

negative of the strictly lower triangular part.

Remember that A we split into D minus L minus L transpose.

So this becomes your M. And your N becomes L. Okay?

So this iteration we say take D (plus, sorry) minus L , well

that's actually l

6:33 / 6:33

2.0x

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Reading assignment

1/1 point (graded)

Read Unit 7.3.2 of the notes. [LINK]

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Jacobi vs Gauss Seidel file missing 3

Homework 7.3.2.1

1/1 point (graded)
Modify the code for (what you now know as the Jacobi iteration) to implement the Gauss-Seidel iteration.

- [x] done

✓

Solution:

For complete solution see the notes .

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Answers are displayed within the problem

Homework 7.3.2.2

1/1 point (graded)
Here we repeat Equation (7.1.1) for Jacobi's iteration applied to the example in Unit 7.1.1:

$$
\left(\begin{array}{cccc|cccc|cc}
4 & & & & & & & & & \\
 & 4 & & & & & & & & \\
 & & 4 & & & & & & & \\
 & & & 4 & & & & & & \\
\hline
 & & & & 4 & & & & & \\
 & & & & & 4 & & & & \\
 & & & & & & 4 & & & \\
 & & & & & & & 4 & & \\
\hline
 & & & & & & & & 4 & \\
 & & & & & & & & & \ddots
\end{array}\right)
\left(\begin{array}{c}
\upsilon_0^{(k+1)} \\ \upsilon_1^{(k+1)} \\ \upsilon_2^{(k+1)} \\ \upsilon_3^{(k+1)} \\ \hline
\upsilon_4^{(k+1)} \\ \upsilon_5^{(k+1)} \\ \upsilon_6^{(k+1)} \\ \upsilon_7^{(k+1)} \\ \hline
\upsilon_8^{(k+1)} \\ \vdots
\end{array}\right)
$$

$$
= \left(\begin{array}{cccc|cccc|cc}
0 & 1 & & & 1 & & & & & \\
1 & 0 & 1 & & & 1 & & & & \\
 & 1 & 0 & 1 & & & 1 & & & \\
 & & 1 & 0 & & & & 1 & & \\
\hline
1 & & & & 0 & 1 & & & 1 & \\
 & 1 & & & 1 & 0 & 1 & & & \ddots \\
 & & 1 & & & 1 & 0 & 1 & & \\
 & & & 1 & & & 1 & 0 & & \\
\hline
 & & & & 1 & & & & 0 & \ddots \\
 & & & & & & \ddots & & \ddots & \ddots
\end{array}\right)
\left(\begin{array}{c}
\upsilon_0^{(k)} \\ \upsilon_1^{(k)} \\ \upsilon_2^{(k)} \\ \upsilon_3^{(k)} \\ \hline
\upsilon_4^{(k)} \\ \upsilon_5^{(k)} \\ \upsilon_6^{(k)} \\ \upsilon_7^{(k)} \\ \hline
\upsilon_8^{(k)} \\ \vdots
\end{array}\right) +
$$

Modify this to reflect the Gauss-Seidel iteration.

- [x] done

✓

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✔ Correct (1/1 point)

Homework 7.3.2.3

1/1 point (graded)

When the Gauss-Seidel iteration is used to solve $Ax = y$, where $A \in \mathbb{R}^{n \times n}$, it computes entries of $x^{(k+1)}$ in the forward order $\chi_0^{(k+1)}, \chi_1^{(k+1)}, \ldots$. If $A = D - L - L^T$, this is captured by

$$(D - L)\, x^{(k+1)} = L^T x^{(k)} + y.$$

Which of the below equations captures a "reverse" Gauss-Seidel method that updates the entries of in the n order $\chi_{n-1}^{(k+1)}, \chi_{n-2}^{(k+1)}, \ldots$.

- ( ) $(D - L)\, x^{(k+1)} = Lx^{(k)} + y$
- (•) $(D - L^T)\, x^{(k+1)} = Lx^{(k)} + y$
- ( ) $(D - L^T)\, x^{(k+1)} = L^T x^{(k)} + y$

✔

Solution:

For complete solution see the notes .

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ℹ Answers are displayed within the problem

Homework 7.3.2.4

1/1 point (graded)

A "symmetric" Gauss-Seidel iteration to solve symmetric $Ax = y$, where $A \in \mathbb{R}^{n \times n}$, alternates between computing entries in forward and reverse order. In other words, if $A = M_F - N_F$ for the forward Gauss-Seidel method and $A = M_R - N_R$ for the reverse Gauss-Seidel method, then

$$\begin{array}{rcl} M_F x^{(k+\frac{1}{2})} & = & N_F x^{(k)} + y \\ M_R x^{(k+1)} & = & N_R x^{(k+\frac{1}{2})} + y \end{array}$$

constitutes one iteration of this symmetric Gauss-Seidel iteration. Determine M and N such that

$$Mx^{(k+1)} = Nx^{(k)} + y$$

equals one iteration of the symmetric Gauss-Seidel iteration.

(You may want to follow the hint in the notes...)

- (•) $\underbrace{(D - L)\, D^{-1}\, (D - L^T)}_{M} x^{(k+1)} = \underbrace{(D - L)\, D^{-1} L(D - \ldots}_{N}$
- ( ) $\underbrace{(D - L^T)\, D^{-1}\, (D - L)}_{M} x^{(k+1)} = \underbrace{(D - L^T)\, D^{-1} L(D - \ldots}_{N}$

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